

GROCERYsim ABM

Scenario Walkthrough Manual

Climate Change & Dairy Supply Chain Disruption

Using the 3D XR-Based Simulator to Explore Consumer Behaviour Under Price Pressure

Finland — Dairy Supply Chain Case Study

SecureFood Project | Horizon Europe | IAMO XR Lab
GROCERYsim ABM v2.0

1. Purpose of This Manual

This manual guides users step-by-step through a specific scenario in the GROCERYsim ABM (Agent-Based Model) application. By following these instructions you will be able to reproduce a simulation that models the downstream effects of climate change on the Finnish dairy supply chain and observe how rising milk and dairy product prices alter consumer purchasing behaviour.

Scenario Context

Climate change has led to more frequent and intense weather events that disrupt the milk and dairy supply chain. The Operational Dashboard (OD) continuously monitors meteorological data (daily weather forecasts, heat stress, droughts, and snow) and shares it with supply chain stakeholders to support preparedness and response.

Digital Twin (DT) simulations indicate that disruption of feed production is expected to drive up feed costs, which in turn increases milk and dairy product prices. The 3D XR-based simulator — GROCERYsim ABM — informs stakeholders about the effects of these price increases on consumer behaviour, including purchasing patterns and demand shifts.

After completing this walkthrough you will have run a paired Baseline vs. Crisis simulation and be able to read and interpret the resulting charts, metrics, and stakeholder dashboards.

2. Prerequisites

Before starting, ensure the following:

- The GROCERYsim ABM application is running and accessible in your web browser.
- You have access to the Finland — Dairy Supply Chain case study data files (Firebase consumer profiles JSON and the dairy products CSV/JSON).
- No prior simulation has been run in the current session (or refresh the page to start fresh).

Tip

If you do not have pre-existing data files, the application will auto-load built-in demo data for the Finland — Dairy Supply Chain case study when you launch it. You can proceed directly to the parameter configuration steps.

3. Step-by-Step Walkthrough

STEP 1

Launch the Application & Select Language

Landing Page

Open the GROCERYsim ABM application in your browser. The landing page will be displayed.

1. **Locate the language switcher** in the top-right corner of the header. The available languages are EN (English), FI (Finnish), EL (Greek), and PT (Portuguese).
2. **Click on your preferred language** button. For this walkthrough, select **EN** (English) unless you require a different language.
3. The page content will immediately update to the selected language.

Note

The language selection affects all labels, descriptions, and button text across the landing page and case studies hub. Simulation parameters and results remain language-independent.

STEP 2

Navigate to Case Studies

Landing Page → Case Studies Hub

From the landing page, access the case study selector:

4. Scroll down slightly until you see the primary call-to-action button labelled **"Explore case studies"**.
5. **Click "Explore case studies"**. The page will transition to the Case Studies Hub.
6. The Case Studies Hub displays four scenario cards, each representing a different market context.

STEP 3

Select the Finland — Dairy Supply Chain Case Study

Case Studies Hub → Simulation

Identify and launch the correct case study:

7. Locate the card titled **"Finland — Dairy Supply Chain"**. Its description reads: *"Simulate dairy product availability, panic-buying dynamics and supply disruptions in the Finnish grocery market."*
8. **Click the "Launch simulation" button** on that card. The simulation workspace will open, replacing the case studies hub.
9. The simulation workspace is organised into a set of tabs across the top: **Data & Population, Interactive Demo, Scientific Analysis, Food Waste, Per-Product, Policy Analysis, Stakeholder View, Sensitivity Analysis, Behavioural Theory, and Export**. You will work through several of these tabs in this walkthrough.

Note

On the left side of the workspace you will find the Simulation Parameters sidebar (collapsed by default on mobile). All scenario parameters are configured here before running the simulation.

STEP 4**Load Data in the Data & Population Tab**

Tab:  Data & Population

The simulation requires consumer profile data and product data. Navigate to the Data & Population tab:

10. **Click on the "Data & Population" tab** (the first tab, labelled  Data & Population).
11. If demo data was auto-loaded when the case study launched, you will see a population summary with archetype percentages and a product list. You can proceed directly to Step 5.
12. If no data is shown, you will see two upload widgets:
 - **Firestore Consumer Profiles (JSON):** Upload the JSON file exported from your Firestore Realtime Database containing consumer demographic and behavioural data.
 - **Product Data (CSV or JSON):** Upload the dairy product catalogue file containing SKUs, prices, fat content, shelf life, and origin (Finnish / imported).
13. After uploading both files, click **"Process Data"**. The system will parse consumer profiles, run K-Means clustering, and assign consumers to behavioural archetypes.
14. Review the population summary that appears. You should see four consumer archetypes with their proportions:
 - **Price Champion** () — maximises value; readily substitutes brands when prices rise.
 - **Green Buyer** () — prefers organic and domestic products; resistant to substitution.
 - **Health Optimizer** () — focuses on nutritional content; moderate price sensitivity.
 - **Habitual Buyer** () — brand-loyal; highest panic-hoarding tendency.

What to look for

A realistic Finnish dairy market profile typically shows a mix dominated by Habitual Buyers and Price Champions. Note the proportions now — they will help you interpret which consumer segments drive demand shifts when prices increase.

STEP 5**Configure Simulation Parameters in the Sidebar**

Sidebar:  Simulation Parameters

The sidebar on the left contains all the levers needed to represent the climate change disruption scenario. Open the sidebar (click the  icon or expand it) and configure each section as described below.

5a. General Settings

Under the  **General** section, set the following:

Parameter	Recommended Value	Rationale
Duration (Days)	365	A full year captures seasonal variation in feed availability and dairy demand, including summer heat stress and winter cold snaps.
Start Month	January (1)	Starting in January aligns the simulation with the agricultural calendar when feed stocks from the previous harvest are being consumed.
Base Daily Consumers	300–500	This calibrates the store as a Medium-to-Large supermarket, representative of a Finnish urban grocery store. The app auto-labels the store tier.

Store Tier Auto-Calibration

300 consumers/day = Medium supermarket (🏪 200–499/day).

500 consumers/day = Large hypermarket (🏬 500–1499/day).

The simulation automatically scales shelf capacity, storage, and stock levels to the selected tier. No manual product-level configuration is required.

5b. Logistics Settings

Under the 🚚 Logistics section, set:

Parameter	Recommended Value	Rationale
Reorder Point (% of max storage)	30%	Triggers restocking when storage falls to 30% capacity — a realistic safety threshold for perishable dairy products.
Restock Target (% of max storage)	90%	Restores storage to 90% upon delivery, providing a buffer against demand spikes.
Lead Time (Days)	3–5 days	Represents realistic supply chain lead time for Finnish domestic dairy and imported products. Increase to 7+ days to simulate disrupted logistics.

5c. Crisis Scenario Settings

This is the core of the climate change scenario. Under the  Crisis Scenario section, configure the parameters that model the downstream impact of weather events on feed supply costs and, consequently, dairy product prices:

Parameter	Recommended Value	Rationale
Crisis Start Day	60–90	Allows a baseline period of 2–3 months before the weather disruption takes effect, mimicking a spring drought onset after winter feed stocks are depleted.
Price Inflation (%)	25–40%	Represents the feed cost pass-through to retail dairy prices. A 25% increase is conservative; 40% models a severe season. Adjust based on DT simulation outputs from Step 4.
Supply Disruption (Days)	10–21 days	Models the period during which supply deliveries are partially or fully disrupted due to weather-related logistics or production failures.
Crisis Duration (Days)	90–180	Set to 0 for an indefinite crisis running to end of simulation, or specify a duration (e.g. 120 days) to observe the full recovery arc after the disruption resolves.

Connecting to the DT Simulation (Step 4)

The Digital Twin simulations referenced in the scenario narrative are expected to quantify how much feed disruption translates into price increases. Use those outputs to inform the Price Inflation (%) parameter above. For example, if the DT predicts a 30% feed cost increase with a 60% dairy price pass-through, set Price Inflation to approximately $30 \times 0.6 = 18\%$ (or higher to represent full pass-through at retail).

5d. Consumer Behaviour Settings

Under the  Consumer Behaviour section:

Parameter	Recommended Value	Rationale
Panic Sensitivity	0.50–0.65	A medium-high value reflects consumers' awareness of weather events and media coverage of supply issues. Increase to 0.75+ for high-media-coverage scenarios.

Parameter	Recommended Value	Rationale
Hoarding Factor	1.5–2.0	Represents the tendency to buy more than usual when a crisis is anticipated. Dairy products (especially long-life milk) are typical panic-hoarding targets.

5e. Policy Scenarios — Domestic Supply Shock

To model the disruption of Finnish domestic dairy feed production specifically (as opposed to a general price shock), enable the Domestic Supply Shock lever. Expand the  Policy Scenarios section in the sidebar, then open the  Domestic Supply Shock sub-panel:

15. Check "Enable Supply Shock" to activate the lever.
16. Set **Shock Start Day** to the same day as your Crisis Start Day (e.g., Day 60–90), so both price and supply disruptions begin simultaneously.
17. Set **Shock Duration (Days)** to 30–60 days, representing the period during which domestic Finnish dairy deliveries are reduced.
18. Set **Shock Severity (fraction blocked)** to 0.50–0.70 (50–70%). This means 50–70% of domestic Finnish deliveries are blocked during the shock period, forcing greater reliance on imported dairy products at higher cost.

What the Supply Shock Models

The Domestic Supply Shock specifically targets Finnish-origin dairy products. When active, the simulation reduces domestic deliveries by the specified severity fraction for the shock duration. Consumers who prefer Finnish dairy (e.g., Green Buyers with high Finnish Preference scores) will be most affected and may shift to imported alternatives or reduce purchases altogether.

STEP 6**Run the Interactive Demo Simulation***Tab:  Interactive Demo*

With all parameters configured, you are ready to run the simulation. The Interactive Demo runs a single paired simulation: one Baseline run (no crisis) and one Crisis run (with the parameters you just set).

19. **Click on the " Interactive Demo" tab.**
20. Optionally adjust the **Animation Speed** slider at the top of the tab. The default is 0.02 seconds per simulated day. Set it to 0.0 for maximum speed or 0.1–0.2 for a slower, more visible progression.
21. **Click the " Run Simulation" button.** The simulation will begin running day-by-day, updating a live revenue chart in real time.
22. Watch the chart as it runs. You should observe:
 - A stable baseline revenue line during the pre-crisis period.
 - A visible divergence between Baseline and Crisis lines starting on the Crisis Start Day.
 - A revenue dip in the Crisis run as demand falls in response to higher prices.
 - Possible recovery after the Crisis Duration ends (if you set a finite duration).
23. Wait for the message **"Simulation complete — explore the other tabs for detailed analysis"** to appear.

STEP 7**Explore the Results Tabs***Interactive Demo → Sub-tabs*

Once the simulation completes, six analysis sub-tabs appear below the live chart. Work through each one to understand the full impact of the dairy price shock on consumers and the store.

7a. Financials

This tab shows revenue, nominal revenue (inflated prices), and average product price over time.

- Constant-price Revenue: falls when consumers buy less — the demand signal.
- Nominal Revenue: rises as prices inflate, even though volume falls. This gap visualises revenue–volume divergence.
- Average Price chart: confirms that price inflation was correctly applied on the Crisis Start Day.

7b. Footfall

Shows the number of consumers visiting the store each day, split by scenario.

- A drop in footfall during the crisis indicates that some consumers are avoiding the store (demand destruction) rather than simply buying smaller baskets.

7c. Logistics

Displays stock levels, orders placed, and deliveries received by the supply truck.

- During the supply disruption period, you should see fewer or delayed deliveries for Finnish dairy products.
- Increased stockouts (shelf going to zero) will appear as gaps or flat lines at the bottom of stock charts.

7d. Crisis Breakdown**Provides a detailed breakdown of the crisis period metrics compared to the baseline period.**

- Revenue impact: the percentage fall in demand-driven revenue.
- Waste change: if consumers stop buying perishable dairy, waste may increase.
- Lost sales: units that consumers wanted but could not purchase due to stockouts.

7e. Behavioural Drift**Shows how consumer preferences shift during the crisis through adaptive learning.**

- Price sensitivity score: rises as consumers experience higher prices.
- Finnish product preference: may decline if Finnish dairy is unavailable and consumers are forced to substitute with imported products.
- Organic preference: may shift depending on which products remain on the shelf.

7f. By Buyer Type**Breaks down all metrics by the four consumer archetypes.**

- Price Champions: likely show the earliest and strongest demand response to price increases, substituting freely.
- Habitual Buyers: show the highest hoarding behaviour early in the crisis.
- Green Buyers: most affected by the domestic supply shock as Finnish dairy disappears from shelves.
- Health Optimizers: moderate response; focus on nutritional alternatives.

STEP 8**Run a Policy Comparison in the Policy Analysis Tab**Tab:  Policy Analysis

The Policy Analysis tab enables you to compare a Baseline run against one or more policy scenarios side-by-side. This is useful for testing interventions such as price subsidies or purchase limits.

24. Click on the " Policy Analysis" tab.
25. In the sidebar, navigate to **Policy Scenarios** and confirm that the Domestic Supply Shock is still enabled (from Step 5e).
26. Optionally, also enable a **Domestic/Organic Subsidy** (e.g. 15% subsidy on Finnish dairy) to simulate a government intervention offsetting the price shock.
27. In the Policy Analysis tab, name your scenario in the "**Scenario name**" field (e.g., "*Supply Shock + 15% Subsidy*").
28. Set **Runs / scenario** to 5 (or more for tighter confidence intervals).
29. Click " **Run & Compare**". The model runs the Baseline and Policy scenarios in parallel across all runs and averages the results.
30. Review the resulting charts showing Revenue, Food Waste, Average Price, and Consumer Welfare metrics under Baseline vs. Policy.
31. To add more scenarios (up to 4), change the sidebar settings, rename the scenario, and click **Run & Compare** again. All saved scenarios will be plotted together for direct comparison.

Interpretation Tip

If the subsidy scenario shows lower average price but similar or higher revenue compared to the supply shock alone, it indicates that the subsidy is successfully maintaining demand volume despite the underlying supply disruption. A divergence in waste levels between scenarios reveals whether the subsidy helps or hinders perishable stock management.

STEP 9**View the Stakeholder Dashboard**Tab:  *Stakeholder View*

The Stakeholder View filters the simulation results to the metrics most relevant for three distinct audiences. This tab is populated from the Interactive Demo simulation you ran in Step 6.

32. **Click on the " Stakeholder View" tab.**
33. At the top of the tab, select your role from the radio buttons:
 - **Retailer** (): focus on revenue, gross margin, waste cost, stockout losses, and shelf availability.
 - **Policy Maker** (): focus on consumer welfare metrics, equity impacts across buyer types, and market stability indicators.
 - **Researcher** (): full access to all metrics including behavioural drift, archetype-level breakdowns, and CO₂ footprint changes.
34. Select the **Crisis** scenario from the Scenario dropdown to see the disrupted market view.
35. Then select **Baseline** to compare the undisrupted market baseline.

For Policy Makers

The Policy Maker view highlights which consumer segments experience the greatest welfare impact from price increases. This directly supports the scenario narrative: the OD shares market fluctuation data with stakeholders to enable evidence-based policy responses to weather-driven supply disruptions.

STEP 10**Export Simulation Data**Tab:  *Export*

All simulation data can be downloaded as CSV files for further analysis or reporting.

36. **Click on the " Export" tab.**
37. You will see download buttons for the following datasets:
 - **Daily Aggregate Data:** Revenue, sales, waste, lost sales, footfall, and average price per day per scenario.
 - **Per-Product Stock Data:** Daily stock levels, sales, waste, and revenue broken down by individual SKU.
 - **Consumer Waste Data:** Waste events categorised by product, reason (expiry, spoilage), and scenario.
38. **Click each "Download CSV" button** to save the corresponding dataset to your computer.
39. These CSV files can be imported into Excel, R, Python, or any statistical analysis tool for further processing, visualisation, or inclusion in reports.

4. Interpreting the Results

The table below summarises the key metrics to look for when interpreting the climate change / dairy supply disruption scenario.

Metric	Expected Direction Under Scenario	Implication
Constant-price Revenue	↓ Decreases	Consumers buy less dairy at higher prices — demand destruction.
Nominal Revenue	↑ Increases initially, then may fall	Price inflation boosts cash flow temporarily; prolonged crisis erodes volume.
Average Basket Size	↓ Decreases	Consumers trade down or substitute cheaper alternatives.
Footfall	↓ Slight decrease	Some consumers reduce visit frequency; high-value items deferred.
Panic Hoarding	↑ Increases at crisis onset	Habitual buyers and others stock up on long-life dairy at first.
Finnish Product Sales	↓ Decreases sharply	Domestic supply shock removes Finnish dairy from shelves.
Imported Product Sales	↑ Increases temporarily	Consumers substitute to available imports; Green Buyers may resist.
Food Waste (perishables)	↑ or ↓ depending on hoarding	Initial hoarding may reduce waste; panic-buying of perishables may increase it later.
Price Sensitivity Score	↑ Increases over time	Consumers adapt preferences toward price-conscious behaviour (behavioural drift).
Green Buyer Purchases	↓ Largest decline	Finnish/organic preference clashes directly with the supply shock.

5. Quick Reference: Recommended Parameter Values

Use this table as a quick reference when setting up the scenario:

Parameter	Recommended Setting	Notes
Duration (Days)	365	Full year for seasonal coverage
Start Month	January	Aligns with agricultural feed cycle
Base Daily Consumers	300–500	Medium to Large supermarket tier
Lead Time (Days)	3–5 (crisis: 7+)	Increase to model disrupted logistics
Reorder Point	30%	Standard dairy safety buffer
Restock Target	90%	Full replenishment target
Crisis Start Day	60–90	After baseline period of 2–3 months
Price Inflation (%)	25–40%	Feed cost pass-through to retail
Supply Disruption (Days)	10–21	Weather-related logistics disruption
Crisis Duration (Days)	90–180 (or 0)	Set 0 for indefinite crisis
Panic Sensitivity	0.50–0.65	Moderate-high awareness level
Hoarding Factor	1.5–2.0	Dairy hoarding tendency
Supply Shock: Enable	Yes	Domestic Supply Shock lever
Supply Shock Start Day	Same as Crisis Start	Simultaneous onset
Supply Shock Duration	30–60 days	Duration of feed disruption
Supply Shock Severity	50–70%	Fraction of Finnish deliveries blocked

6. Troubleshooting

No data loaded warning

If you see a yellow warning banner saying "Load data in the Data & Population tab first", return to the Data & Population tab and either upload your data files or confirm that the auto-loaded demo data is visible. Do not proceed to run a simulation until this is resolved.

Simulation results look identical in Baseline and Crisis

Verify that Price Inflation is set above 0% and Crisis Start Day is less than Duration (Days). If Crisis Start Day equals or exceeds Duration, the crisis never begins within the simulation window.

Policy Analysis tab shows no active policies

The Policy Analysis tab reads policy settings from the sidebar. Ensure you have checked "Enable Supply Shock" in the Domestic Supply Shock expander before clicking Run & Compare.

Stakeholder View tab is empty

The Stakeholder View is populated from the Interactive Demo simulation. If you have not yet run the demo (Step 6), the Stakeholder View will show a warning. Run the Interactive Demo first.

— *End of Walkthrough Manual* —